LOKESH KUMAR (SONARA)

Curriculum Vitae — Europass Format | Software Engineer & Applied AI Researcher

Date of Birth: 18/08/2003 | Gender: Male | Nationality: Pakistani
Passport No: AA8310592 (Valid thru 25/04/2033) | National Identity (CNIC): 44303-2907059-9
Place of Birth: Tharparkar, Sindh, Pakistan | Father's Name: Santosh Kumar
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IEEE Student Member: #99402233 (IEEE Karachi Section) | ORCID iD: https://orcid.org/0009-0006-6510-6393
Hugging Face (Open-Source AI Datasets): https://huggingface.co/datasets/lokeshsonara/dhatki-agri-intents

BRIEF PROFILE & RESEARCH OBJECTIVE

Final-year Software Engineering researcher and developer at Sindh Agriculture University (SAU) Tando Jam, specializing in applied AI, edge computing systems, and data pipelines. Lead Developer and proposal author for the $10,000 EPICS in IEEE grant initiative for Agri-Logistics IDAS under IEEE Karachi Section. Proven track record across 3 industry internships and 90+ deployed software architectures. Seeking admission to competitive European Master's programs (Erasmus Mundus SE4GD / MULTICOM) to pioneer sustainable, localized, and socio-technical software solutions bridging human needs with frugal computing.

EDUCATION AND TRAINING

Bachelor of Science in Software Engineering (BS-SE)
Information Technology Centre, Sindh Agriculture University (SAU), Tando Jam, Pakistan | 2022 – Expected Dec 2026
- Cumulative GPA: 3.00 / 4.00 (Peak Semester 6 Academic Distinction GPA: 3.65 / 4.00, Grade A-)
- Core Engineering Coursework: Data Structures & Algorithms, Artificial Intelligence & Machine Learning, Database Management Systems, Software Architecture & Design, Operating Systems, Computer Networks, Web Engineering, Software Project Management.
- Medium of Instruction: English (Certified by Official University Notification No. ENG 375).

Higher Secondary School Certificate (HSC-II) — Pre-Medical
Govt. Boys Degree College Mithi, Tharparkar (BISE Mirpurkhas) | 2021 – 2023
- Academic Standing: Grade A-1 (Top Tier Distinction: 94.18% — 1036 / 1100 Total Marks).

Secondary School Certificate (SSC-II) — Science
Amar Jagdesh Malani Govt. High School North Colony Mithi, Tharparkar (BISE Mirpurkhas) | 2018 – 2020
- Academic Standing: Grade A (71.88% — 611 / 850 Total Marks).

PUBLICATIONS & OPEN-SOURCE AI DATASETS

L. Kumar and B. S. Chowdhry, "Bridging the Digital Literacy Gap in Rural Agri-Logistics: A Trilingual, Context-Aware Intelligent Driver Assistance System (Agri-Logistics IDAS)," SAU & MUET Joint Research Preprint, Oct. 2026.
Live System Portal: https://agri-idas.tech/ | Full Paper PDF: https://agri-idas.tech/AgriLogistics_IDAS_IEEE_TechRxiv_Paper.pdf | ORCID: 0009-0006-6510-6393
- Demonstrated across 8 Lower Sindh transit corridors (1,313 OSRM road-centerline points) that straight-line Haversine routing induces a 17.07% distance underestimation error (t=4.892, p=0.0017), causing +8.84 L of unbudgeted diesel burn (Rs. 2,488.81 PKR) per trip.
- Engineered a phonetic dialect NLP engine achieving Dhatki semantic intent classification in 0.0034 ms and ISO 26262 compliant auditory voice synthesis in 747.92 ms.

Dhatki & Sindhi Vernacular Agricultural Logistics Intent Corpus (dhatki-agri-intents) — Open-Source AI Dataset
Published on Hugging Face Hub (MIT License) | URL: https://huggingface.co/datasets/lokeshsonara/dhatki-agri-intents
- Authored and released the first open-source parallel vernacular dataset for under-resourced Dhatki and regional Sindhi agricultural freight intents, enabling edge-AI acoustic and semantic modeling for rural supply chains.

RESEARCH & FLAGSHIP APPLIED ENGINEERING PROJECTS

Agri-Logistics IDAS (Intelligent Decision & Advisory System) — Lead Developer & Proposal Author
International EPICS in IEEE ($10,000 Grant Initiative) in collaboration with IEEE Karachi Section & SPO | 2026
- Spearheaded the complete technical architecture and grant proposal for an offline, vernacular logistics co-pilot tackling post-harvest agricultural transit loss across rural Sindh corridors.
- Engineered offline NLP speech quantization pipelines supporting bidirectional voice interaction in Dhatki (Mother Tongue), Sindhi, and Urdu with sub-second response times on low-cost edge computing hardware.
- Integrated open-source geospatial mapping (NetworkX / Folium) with localized road vibration datasets to optimize heavy freight routing along stable asphalt centerlines.

AI-Powered Virtual Assistant & Voice Automation Hub — Capstone Final Year Project (FYP)
Department of Software Engineering, SAU Tando Jam (Supervised by Madam Sohni Abbasi) | 2026
- Architecting an autonomous desktop-to-mobile workflow automation engine bridging operating system commands and Natural Language Processing (NLP) models.
- Implemented background event-listening services and low-latency API bridges for hands-free local application execution.

NexusOmni (ContractIQ) — AI Legal Contract Risk Scoring Platform
Independent Production Platform (Deployed live on Vercel) | Next.js 15, TypeScript, React
- Engineered an AI document parser evaluating legal liability in service agreements and NDAs, outputting 0-100 risk metrics and red-flag clause extractions under 60 seconds.

S-Jeewlani Jewellers — Real-Time Bullion E-Commerce Platform & Mobile App
Commercial Production System | Next.js 15, REST APIs, Google Play Store App
- Engineered a full-stack digital storefront with live automated Sarafa Association bullion rate synchronization, precise transparent price calculations, and automated WhatsApp rate subscription pipelines.

WORK & INTERNSHIP EXPERIENCE

Summer Intern — Volunteer Management Department
Alkhidmat Foundation Pakistan (Ref: AKFP/VM/2883) | June 2026 – August 2026 (8 Weeks)

- Contributed to large-scale volunteer mobilization workflows, database consolidation, and emergency welfare logistics.
- Awarded formal Certificate of Achievement and Official Completion Letter authorized by Senior Executive Leadership (Verification: https://freeimage.host/i/nniprf1).

Data, Operations & AI Intern

Pakistan Hindu Council | March 2026 – May 2026 (6 Weeks, Hybrid)

- Restructured manual community data collection pipelines into secure, automated Python reporting formats, reducing operational reporting latency by 40% (Verification: https://freeimage.host/i/CBVPY8I).
- Designed privacy-compliant, anonymized database schemas for sensitive regional welfare tracking.

Web Development Intern

Interns Pakistan | December 2024 – January 2025 (4 Weeks)

- Executed front-end interface engineering tasks using HTML5, CSS3, JavaScript, and responsive design frameworks (Verification: https://freeimage.host/i/C26XtQR).

VERIFIED CERTIFICATIONS & ACCREDITATIONS

- DeepLearning.AI: AI For Everyone — Instructed by Prof. Andrew Ng | Credential ID: BTJWB77W6W2Q (Verify: https://coursera.org/account/accomplishments/verify/BTJWB77W6W2Q)
- IBM: Introduction to Software Engineering — SDLC, Agile principles, software architecture | Credential ID: E42GWWF22X5S (Verify: https://coursera.org/verify/E42GWWF22X5S)
- Google: Foundations of Cybersecurity — Threat modeling, network security, risk protocols | Credential ID: 3NZWPQQJBMQ5 (Verify: https://coursera.org/account/accomplishments/verify/3NZWPQQJBMQ5)
- Google: Google Agile Essentials Specialization (3-Course Specialization) | Credential ID: GZW3MCSAGXNX (Verify: https://coursera.org/verify/specialization/GZW3MCSAGXNX)
- University of Michigan: Programming for Everybody (Python) | Verified Credential (Verify: https://coursera.org/share/8eac2c1e01e3d75b1554530245d50150)
- World Bank Group: Sustainable MSME Finance — Financial mechanisms & ESG compliance | Verified Credential (Verify: https://freeimage.host/i/CgjXDrX)
- Nestlé Enterprise E-Learning Sprint (NESTGEN 2026 — 16 Verified Credentials): Covering Industry 4.0, Circular Economy & ESG, AI Ethics, Supply Chains, and Cloud SaaS | Verified Portfolio (Verify: https://drive.google.com/file/d/1A54UrK9jcSI5ezQpCR_5QeqLapoYbIVc/view?usp=sharing)
- IEEE Karachi Section SAC & IEEE Education Society: Technical Talk Series Certificate (Speaker: Prof. Dr. Bhawani Shankar Chowdhry) (Verify: https://freeimage.host/i/CpWUxI9)
- IEEE Computer Society: Certificate of Membership 2026 (Verify: https://freeimage.host/i/CmtUsVe)

TECHNICAL SKILLS & COMPETENCES

- Programming Languages: Python (Advanced), JavaScript / TypeScript, SQL (PostgreSQL, MySQL), C++, HTML5, CSS3.
- Frameworks & Web Technologies: Next.js 15, React.js, Node.js, Streamlit, Tailwind CSS, REST APIs.
- Data Engineering & AI: Natural Language Processing (NLP), Speech Quantization, Pandas, NumPy, NetworkX, Folium, Data Anonymization, RAG Systems.
- Cloud, DevOps & Tools: Git, GitHub, Docker, Google Cloud Platform (GCP), Vercel Global CDN, Netlify, Linux/Bash.

LANGUAGE SKILLS (CEFR Standard)

- Dhatki: Mother Tongue (C2 — Native Proficiency).
- Sindhi: Native Regional Language (C2 — Native Proficiency).
- Urdu: National Language (C2 — Bilingual Proficiency).
- English: C1 / B2 (Proficient User) — Medium of Instruction Certified (IELTS Academic Booked).

HONOURS, LEADERSHIP & COMPETITIVE AWARDS

- IEEE Student Member (#99402233) — IEEE Karachi Section & IEEE SAU Student Branch (2024 – Present) (Verify: https://freeimage.host/i/CgahMqN).
- Lead Proposal Author: $10,000 EPICS in IEEE International Community Service Grant Proposal (2026).
- National Talent Award: Great Future Talent Olympiad (Batch VI, Dec 2025) — Awarded for exceptional scholastic performance (Verify: https://freeimage.host/i/C26q5TQ).
- QuizOff 2026 AI Competition (Unstop): Competed in India's Biggest AI Quiz across 525,000+ international participants from 35+ countries.
- Academic Distinction: Scored 94.18% (Grade A-1) in Higher Secondary School Examination (2023).
- Community Mentorship: Mentored junior software engineering undergraduates in Python and Git at SAU Tando Jam.